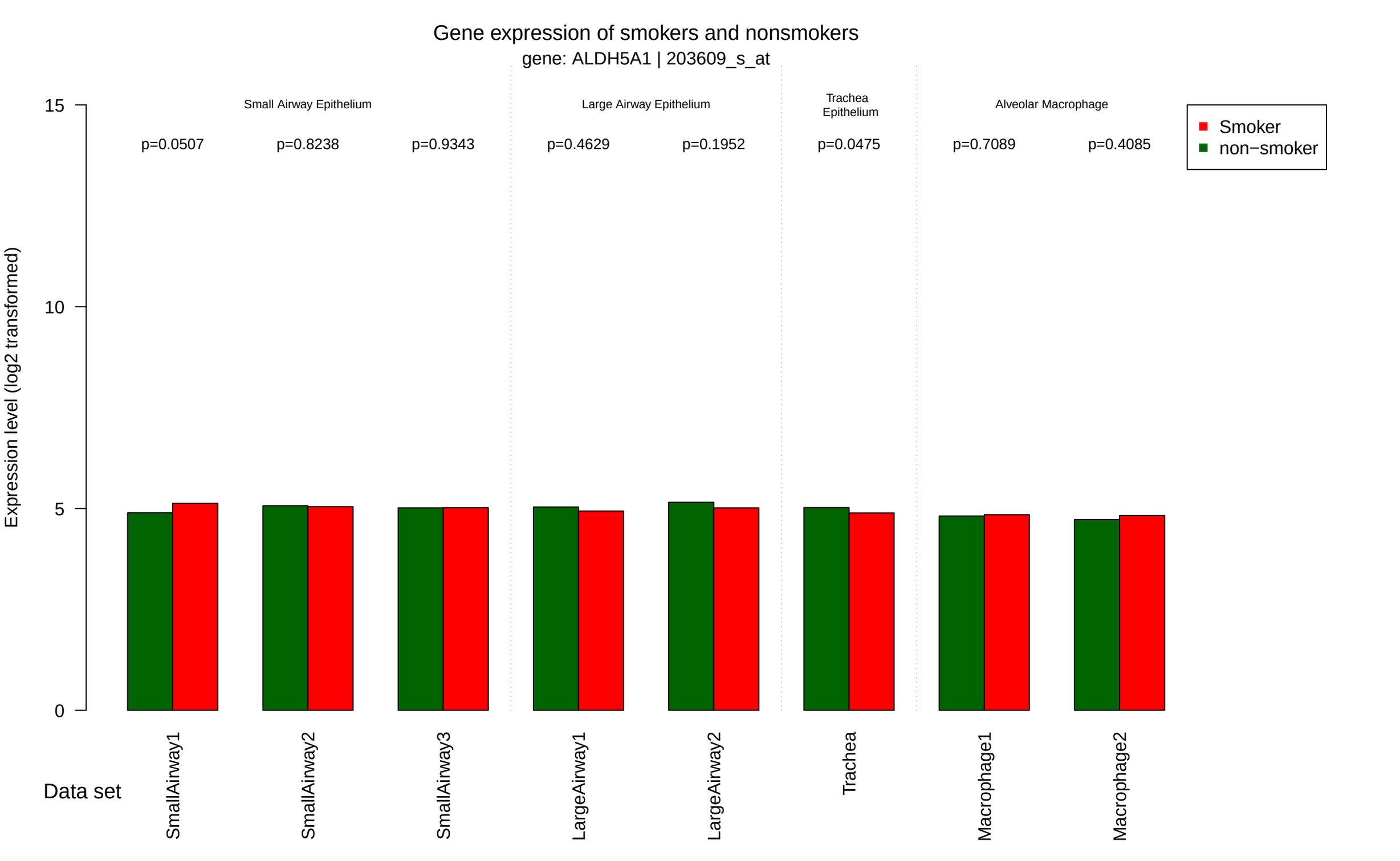
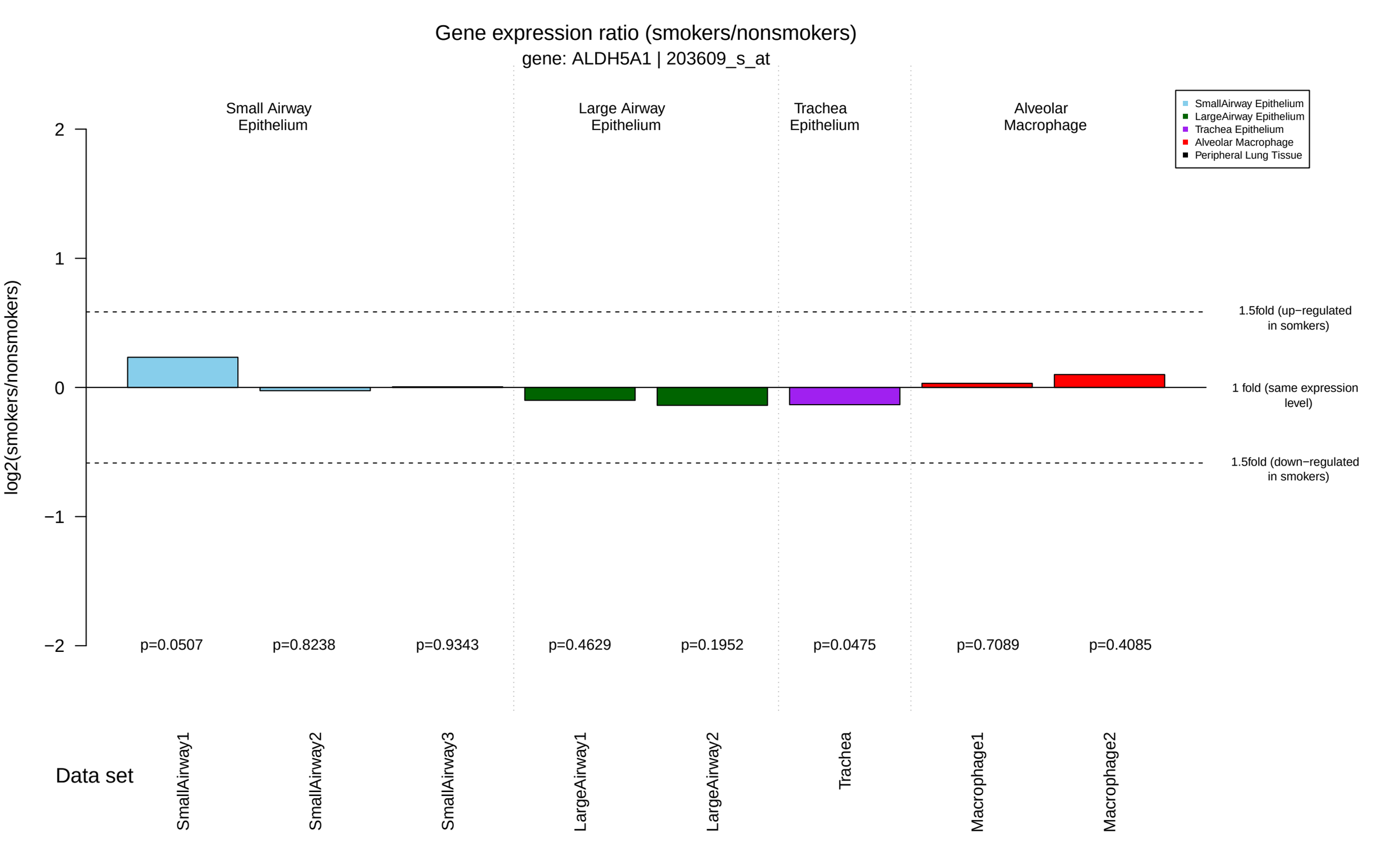
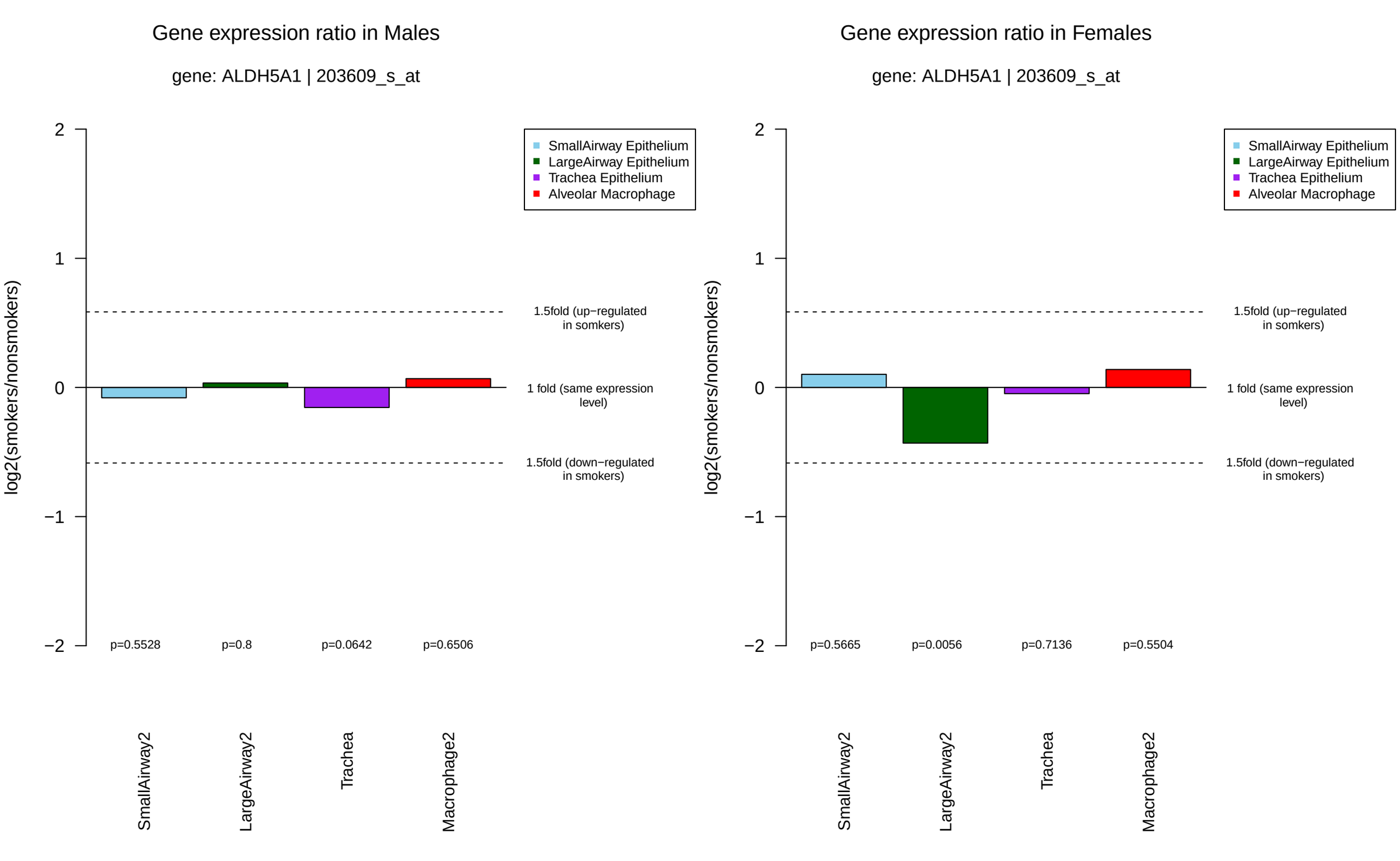
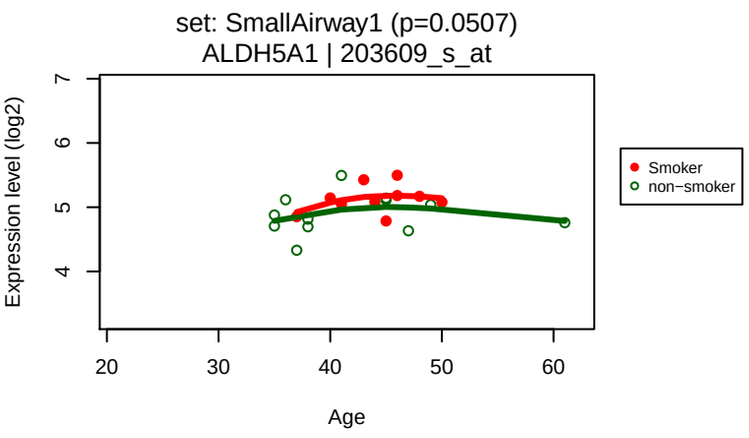
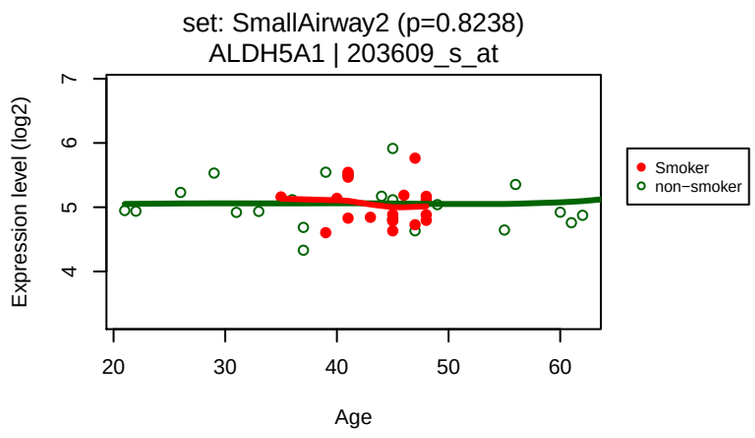
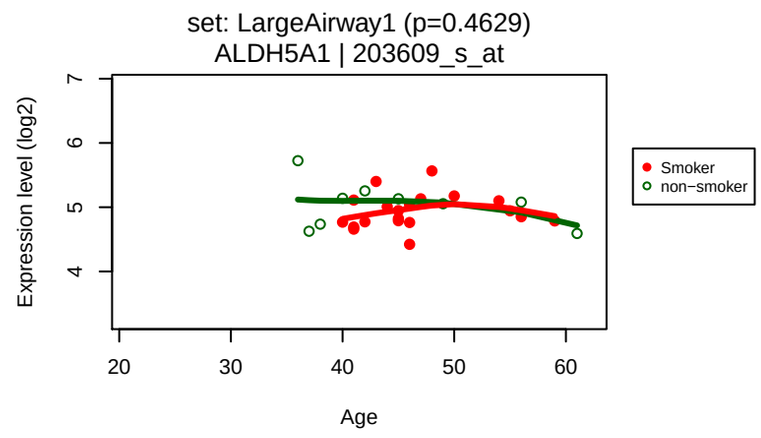

Gene expression of smokers and nonsmokers
gene: ALDH5A1 | 203609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0507
p=0.8238
p=0.9343
p=0.4629
p=0.1952
p=0.0475
p=0.7089
p=0.4085
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

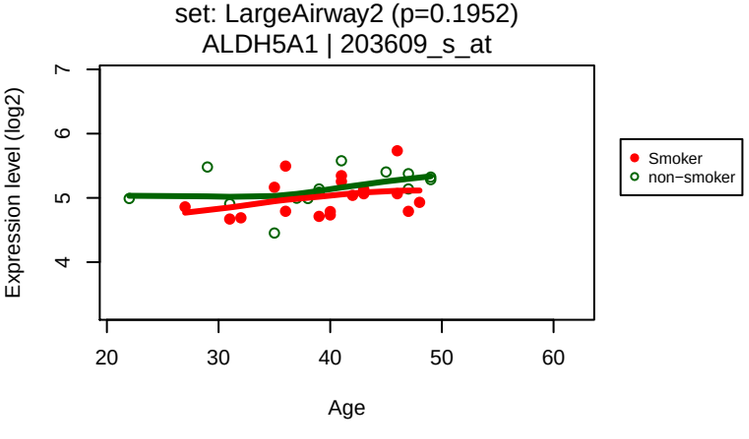

Gene expression ratio (smokers/nonsmokers)
gene: ALDH5A1 | 203609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0507
p=0.8238
p=0.9343
p=0.4629
p=0.1952
p=0.0475
p=0.7089
p=0.4085
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

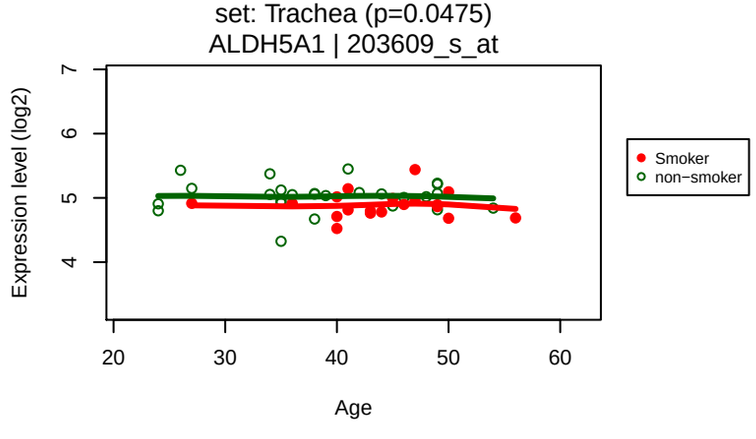

Gene expression ratio in Males
gene: ALDH5A1 | 203609_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5528
p=0.8
p=0.0642
p=0.6506
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH5A1 | 203609_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5665
p=0.0056
p=0.7136
p=0.5504
SmallAirway2
LargeAirway2
Trachea
Macrophage2

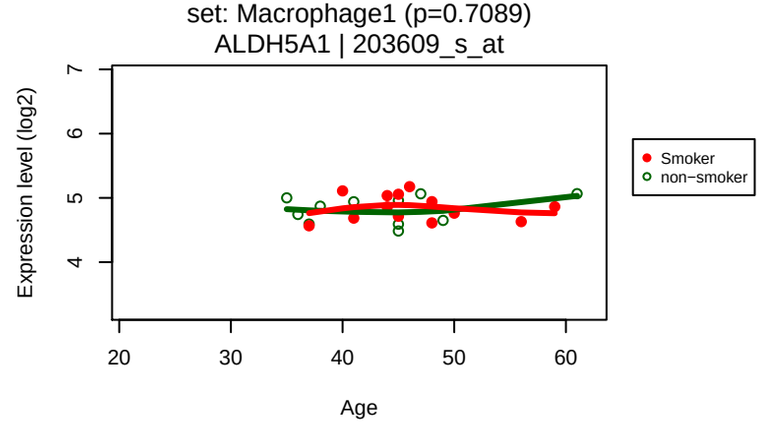

set: SmallAirway1 (p=0.0507)
ALDH5A1 | 203609_s_at
Expression level (log2)
Age
Smoker
non−smoker

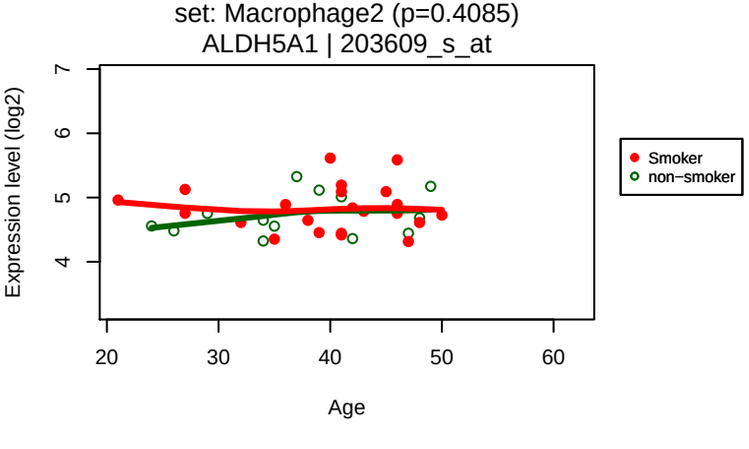

set: SmallAirway2 (p=0.8238)
ALDH5A1 | 203609_s_at
Expression level (log2)
Age
Smoker
non−smoker

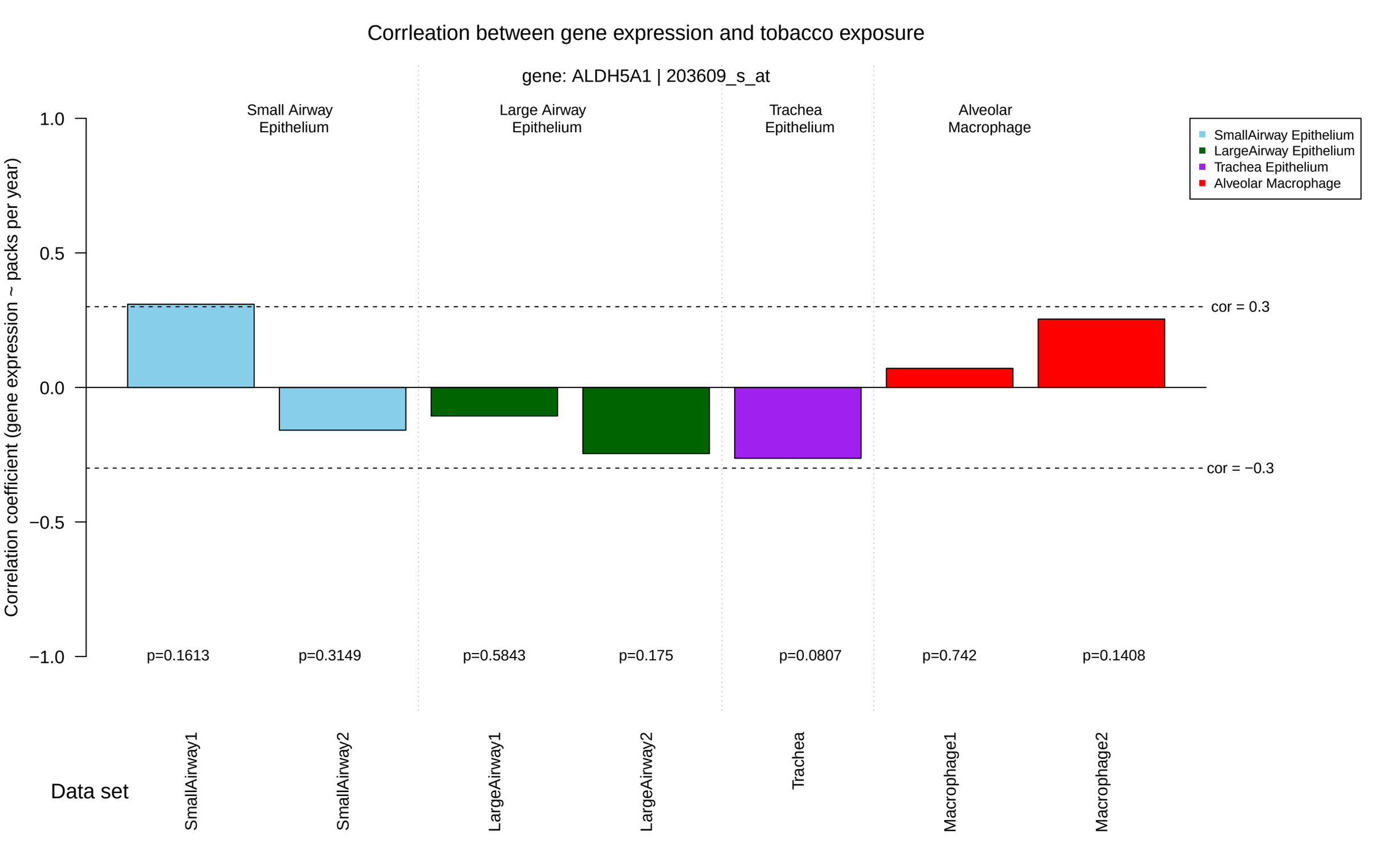

set: LargeAirway1 (p=0.4629)
ALDH5A1 | 203609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1952)
ALDH5A1 | 203609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0475)
ALDH5A1 | 203609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7089)
ALDH5A1 | 203609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4085)
ALDH5A1 | 203609_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH5A1 | 203609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1613
p=0.3149
p=0.5843
p=0.175
p=0.0807
p=0.742
p=0.1408
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2